

Phase 1 — Framing Memo

An adjoint / weight-window theory of neural rare-event samplers
for statistical mechanics

Purpose. This memo executes Phase 1 of the plan: lock the conceptual framing before writing code. It is the theory backbone of the paper. The deliverable is not a vocabulary table but a single rigorous identification — the transport adjoint importance function, the rare-event committor, and the “neural importance function” now being reinvented in machine learning are one object — together with the consequence that the mature CADIS / FW-CADIS / weight-window / figure-of-merit apparatus of deep-penetration transport is the correct and currently-unused home for learned rare-event sampling of spin systems. Where the analogy is exact it is proved; where it is only heuristic it is flagged as such.

1 The one claim worth staking

The literature review found that the mathematical objects we need are already being rediscovered on the machine-learning side (notably Kim & Cai 2026, §6), but without the transport-theoretic scaffolding that has surrounded them since the 1950s. Our defensible contribution is therefore *not* “use a learned importance function to sample rare events” — that now exists. It is the following.

Claim 1 (Framing) *Learned importance sampling of rare events in statistical mechanics is a special case of adjoint-driven variance reduction in linear transport. Casting it there (i) identifies the optimal learned sampler with the transport zero-variance scheme, (ii) supplies the global objective of FW-CADIS — uniform relative error across a whole family of rare observables — which no neural rare-event method currently targets, and (iii) supplies the figure of merit $FOM = 1 / (\sigma_{\text{rel}}^2 T)$ as a training objective, replacing the reverse-KL free energy that is mismatched to rare-event goals.*

2 Two formalisms, stated precisely

2.1 Linear transport and the adjoint importance function

Let $P = (\mathbf{r}, \Omega, E)$ be a phase-space point and $\psi(P)$ the particle flux obeying the linear transport (Boltzmann) equation, written abstractly as

$$\mathcal{H} \psi = q, \tag{1}$$

with transport operator $\mathcal{H}$ (streaming + collision) and source q . A detector response (“tally”) is a linear functional

$$R = \langle \sigma_d, \psi \rangle = \int \sigma_d(P) \psi(P) dP, \tag{2}$$

where σ_d is the detector cross-section. Define the *adjoint* flux $\psi^\dagger$ by

$$\mathcal{H}^\dagger \psi^\dagger = \sigma_d. \tag{3}$$

The defining property of the adjoint — $\langle \mathcal{H}^\dagger \psi^\dagger, \psi \rangle = \langle \psi^\dagger, \mathcal{H} \psi \rangle$ — gives the **fundamental identity**

$$R = \langle \sigma_d, \psi \rangle = \langle \psi^\dagger, q \rangle \quad (4)$$

and the physical reading that makes everything work: $\psi^\dagger(P)$ is the *expected contribution to the tally R of a single particle introduced at P — its importance*. A particle deep inside a shield has tiny importance; one near the detector has large importance.

Zero variance. If one samples particle transitions biased toward high importance and assigns each particle the statistical weight

$$w(P) \propto \frac{1}{\psi^\dagger(P)}, \quad (5)$$

then — when $\psi^\dagger$ is exact — *every* history contributes the identical score and the estimator of R has *zero variance*. This is the theoretical ceiling of variance reduction; all practical schemes approximate it.

2.2 CADIS, FW-CADIS, weight windows, figure of merit

Since $\psi^\dagger$ is unknown, CADIS (Consistent Adjoint Driven Importance Sampling) computes a *cheap, approximate* $\psi^\dagger$ from a fast deterministic solve of (3), then uses it to drive an expensive forward Monte Carlo run through two consistent devices:

- **Source biasing:** sample the source from $\hat{q}(P) = \psi^\dagger(P)q(P)/R$ instead of q ;
- **Weight windows:** give each phase-space cell a target weight (5) and a window $[w_{\text{target}}/k, k w_{\text{target}}]$; a particle above the window is *split* (into equal-weight copies), one below is *Russian-rouletted*. This keeps $w(P) \psi^\dagger(P) \approx \text{const}$, i.e. bounds the weight and hence the variance.

The **consistency condition** is essential and often overlooked: the source biasing and the weight windows must agree, or freshly sampled particles land outside the window and are immediately split/rouletted, wasting the effort. **FW-CADIS** (forward-weighted CADIS) extends this to *global* variance reduction: by weighting the adjoint source with the inverse of the local forward response, it produces weight windows that equalise the *relative* statistical error across many detectors or the whole domain at once, rather than optimising a single tally. Efficiency is measured by the

$$\text{figure of merit} \quad \text{FOM} = \frac{1}{\sigma_{\text{rel}}^2 T}, \quad (6)$$

the inverse product of relative variance and CPU time; variance reduction maximises it.

2.3 Gibbs equilibrium, the VAN, and the rare-event targets

On the statistical-mechanics side the object is the Gibbs measure $p(s) = e^{-\beta E(s)}/Z$, approximated by a variational autoregressive network $q_\theta(s) = \prod_i q_\theta(s_i | s_{<i})$ trained by minimising the reverse-KL variational free energy $\beta F_q = \mathbb{E}_q[\beta E + \ln q_\theta] \geq \beta F_{\text{exact}}$. The quantities we actually care about are *rare*:

- tail probabilities $\pi = \Pr[A(s) \geq a^*]$ of an observable A (e.g. a large magnetisation fluctuation, an interface, a nucleus);
- transition rates between metastable states;
- the large-deviation rate function $I(a) = -\lim_{N \rightarrow \infty} \frac{1}{N} \ln \Pr[A/N \approx a]$.

An honesty note. The loose analogy “VAN training is importance sampling with weight windows” that one is tempted to draw at the level of *static equilibrium sampling* is only heuristic: equilibrium Gibbs sampling has no intrinsic transport operator, so there is no genuine adjoint there. The rigorous bridge runs through *rare events and dynamics*, where a bona-fide adjoint (the committor / tilted generator) exists. That is the subject of the next section, and it is where the novelty is real.

3 The central identity: importance = committor = zero-variance measure

Work with a Markov chain — either a physical dynamics on configurations s , or the sequential “time” of the autoregressive construction — with transition kernel $K(s \rightarrow s')$. Let A (failure / source) and B (the rare target set) be disjoint, and define the **committor**

$$h(s) = \Pr[\text{reach } B \text{ before } A \mid \text{start at } s], \quad h|_A = 0, \quad h|_B = 1. \quad (7)$$

On the transient set $\mathcal{T}$ it satisfies the discrete backward equation

$$h(s) = \sum_{s'} K(s \rightarrow s') h(s') \quad \Longleftrightarrow \quad (\mathbb{I} - K)h = 0 \text{ on } \mathcal{T}. \quad (8)$$

Equation (8) is precisely the discrete *adjoint* transport equation (3) with the target set playing the role of the detector/adjoint source. Hence:

Proposition 1 (Coincidence of the three objects) *The rare-event committor $h(s)$, the transport adjoint importance $\psi^+(s)$ (with $\sigma_d = \mathbf{1}_B$), and the optimal “neural importance function” are the same function: the expected contribution of a walker at s to the rare tally.*

Proposition 2 (Zero-variance change of measure) *Define the tilted kernel*

$$\tilde{K}(s \rightarrow s') = K(s \rightarrow s') \frac{h(s')}{h(s)}. \quad (9)$$

By (8) it is a proper stochastic kernel ($\sum_{s'} \tilde{K} = 1$), it never steps into A (since $h|_A = 0$), and the path likelihood ratio telescopes,

$$W = \prod_k \frac{K(s_k \rightarrow s_{k+1})}{\tilde{K}(s_k \rightarrow s_{k+1})} = \prod_k \frac{h(s_k)}{h(s_{k+1})} = \frac{h(s_0)}{h(s_{\text{end}})} = h(s_0) \quad (10)$$

for every path (all of which end in B , $h = 1$). Every sample returns the identical value $h(s_0) = \pi$, so the estimator of π has zero variance.

The transport weight-window target (5), $w \propto 1/\psi^+ = 1/h$, is exactly the practical, robust approximation to this ideal: rather than requiring the exact h (which would need solving the problem), one bounds $w \cdot h \approx \text{const}$ by splitting and rouletting. Figure 1 demonstrates all of this on the cleanest possible instance — a biased walk toward an absorbing boundary, the one-dimensional caricature of both deep-penetration shielding and a rare fluctuation reaching a target set.

4 The dictionary

Table 1 is the correspondence, with each row marked [E] exact (a theorem-level identity) or [H] heuristic (a useful but non-rigorous analogy).

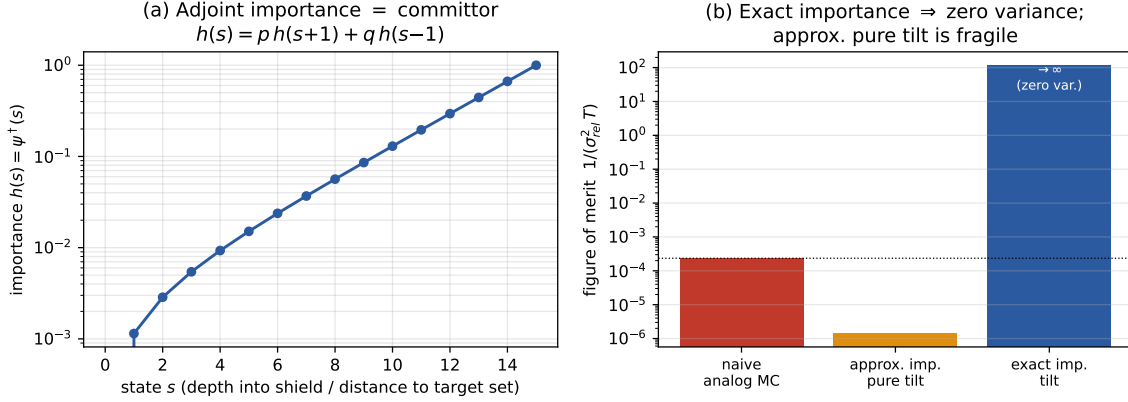


Figure 1: Exact demonstration of §3 (computed). A nearest-neighbour walk on $\{0, \dots, 15\}$ drifts toward 0; reaching 15 first is the rare deep-penetration event, $\pi = 1.14 \times 10^{-3}$. (a) The importance $h(s) = \psi^\dagger(s)$ rises steeply toward the target and obeys the adjoint equation (8). (b) Figure of merit. Naive analog Monte Carlo (red) is the baseline; tilting (9) with the *exact* importance (blue) gives measured relative variance $\sim 10^{-30}$ — zero variance to machine precision; tilting with an *approximate* importance under *pure biasing* (amber) is *worse than doing nothing*. That fragility is exactly why deep-penetration transport uses weight windows (split/roulette) to bound the weights — the robustness layer whose gain on spin systems Phase 2 will quantify.

5 Consequences the ML literature is not using

The FOM is the right objective, not reverse-KL. Reverse-KL trains q_θ to match the *whole* Boltzmann distribution p . But the zero-variance sampler for a rare tail is not p ; it is the tilted measure $p(s)h(s)/\pi$, concentrated on the configurations that realise the event. Training a VAN by minimising the *relative variance* of the tail estimator — equivalently maximising the FOM — targets that tilted measure directly. This is a concrete, implementable change of loss with a transport pedigree, and it sidesteps the known failure of global density matching for rare events.

The consistency condition is a design principle, not an afterthought. The demo’s amber bar is the ML community’s recurring “importance sampling made it worse” failure. In transport it has a name and a fix: keep the proposal (source biasing) consistent with the weight map, and bound weights with windows. Porting this discipline to neural proposals is low-risk and directly useful.

Global variance reduction is untouched. Every neural rare-event method we found optimises *one* event or rate. FW-CADIS’s signature — one importance map giving uniform relative error across a whole family of targets (e.g. the entire large-deviation curve $I(m)$ of the magnetisation) — has no neural counterpart. This is the single most defensible technical gap.

6 Positioning against the two closest works

Kim & Cai, arXiv:2602.12294 (Jan 2026) — the dangerous near-competitor — learns an importance function $I(s)$, modifies transitions as $K'(i \rightarrow j) = K(i \rightarrow j)I(j)/I(i)$, uses a bias potential $E_b = -2k_B T \ln I$, and a branching random walk. In the language above: their I is our $h = \psi^\dagger$; their K' is the tilt (9); their E_b is the weight-window target in energy units; their branching is splitting. They reconstructed the zero-variance/weight-window recipe for discrete rare transitions — *without the*

Table 1: Radiation-transport variance reduction $\leftrightarrow$ neural rare-event sampling.

Transport (deep penetration)	Neural sampler (rare events in stat. mech.)	
Adjoint flux / importance $\psi^\dagger(P)$	Committer $h(s)$; learned importance $I_\theta(s)$	[E]
Adjoint equation $\mathcal{H}^\dagger \psi^\dagger = \sigma_d$	Backward/committer eqn. $(\mathbb{I} - K)h = 0$, target = adjoint source	[E]
Fundamental identity $R = \langle \psi^\dagger, q \rangle$	$\pi = \mathbb{E}_p[\mathbf{1}_B] = h(s_0)$	[E]
Importance-weighted zero-variance scheme	Doob / committer-tilted zero-variance sampler (9)	[E]
Weight-window target $w \propto 1/\psi^\dagger$	Bounded importance weights $w \propto 1/h$	[E]
Splitting / Russian roulette	Branching / resampling of configurations (robustness layer)	[E]
CADIS: cheap adjoint drives forward MC	Cheap/learned I_θ drives the neural proposal	[H]
Source-weight-window consistency condition	Proposal must match the weight map or weights blow up	[E]
FW-CADIS: uniform relative error, all tallies	Uniform error across a family of tails / whole $I(a)$	[H]
Figure of merit $1/(\sigma_{\text{rel}}^2 T)$	Training objective for the sampler	[E]
Static equilibrium flux	Static Gibbs sampling	[H]

transport framing. What they do *not* do, and what remains open: the transport FOM as objective; the *global* FW-CADIS uniform-error goal; application to the spin-system Boltzmann tails; a VAN (autoregressive, exactly-normalised) ansatz; and the consistency condition as a stated design principle.

Doob-transform VAN, Phys. Rev. E 111, 034120 (2025) approximates the leading eigenvector of the *tilted generator* with a VAN for *dynamical* large deviations. That eigenvector is the dynamical analogue of $\psi^\dagger$. What remains open: static rare-event tails; the CADIS/weight-window robustness layer; the FOM objective; and global (multi-target) variance reduction.

7 What survives, and the bridge to Phase 2

Three claims survive the prior-art check and are worth building:

1. **Unification (framing).** Identify $\psi^\dagger = h = I_\theta$ and place the recent neural rare-event methods inside 70-year-old transport variance-reduction theory, importing weight-window generators, the consistency condition, and the FOM wholesale.
2. **Global FW-CADIS for spin systems (headline).** One learned importance map that equalises relative error across a family of magnetisation/energy tail thresholds — the whole rate function $I(m)$ — rather than a single event.
3. **FOM-trained sampler.** Train the VAN/importance map by maximising $1/(\sigma_{\text{rel}}^2 T)$ under the consistency condition, and show robustness where pure tilting (and reverse-KL) fail.

Phase 2 experiments that convert framing into a result. On the 2D Ising model (exact Onsager / Ferdinand–Fisher references available): (i) learn an approximate committor/importance for a rare magnetisation tail; (ii) build FW-CADIS-style global weight windows across a family of thresholds; (iii) measure the FOM against two baselines — a reverse-KL VAN with importance reweighting, and a single-target Kim–Cai-style biasing — with the *global uniform-error* advantage as the headline metric; (iv) repeat on the Edwards–Anderson spin glass to show the weight-window robustness layer survives a rugged landscape (where the PNAS 2024 result shows plain neural samplers struggle). A positive FOM gain of even $2\text{--}5\times$ at fixed cost, *plus* the global-error property, clears *Mach. Learn.: Sci. Technol.* or *Phys. Rev. E*; a demonstrated 3D or spin-glass generalisation reaches for a stronger venue.

Immediate next action. Nothing here needs new code. The single most valuable Phase-2 seed is the exactly-solvable committor demo of Fig. 1, extended one step: add the weight-window (split/roulette) layer and show it recovers robust variance reduction from an *approximate* importance — turning the amber bar green. That is a half-day of work on the 1D model and becomes the controlled unit test for the 2D-Ising implementation.

Key references

1. D. Wu, L. Wang & P. Zhang, *Solving statistical mechanics using variational autoregressive networks*, Phys. Rev. Lett. **122**, 080602 (2019); arXiv:1809.10606.
2. M. Kim & W. Cai, *Accelerated MCMC via neural network-driven importance sampling*, arXiv:2602.12294 (2026). [closest prior art]
3. *Nonequilibrium statistical mechanics revealed by Doob transform and variational autoregressive networks*, Phys. Rev. E **111**, 034120 (2025).
4. J. C. Wagner & A. Haghighat, *Automated variance reduction of Monte Carlo shielding calculations using the discrete ordinates adjoint function (CADIS)*, Nucl. Sci. Eng. **128**, 186–208 (1998).
5. J. C. Wagner, D. E. Peplow & S. W. Mosher, *FW-CADIS method for global and regional variance reduction...*, Nucl. Sci. Eng. **176**, 37–57 (2014).
6. M. A. Cooper & E. W. Larsen, *Automated weight windows for global Monte Carlo particle transport*, Nucl. Sci. Eng. **137**, 1–13 (2001).
7. K. A. Nicoli et al., *Asymptotically unbiased estimation of physical observables with neural samplers*, Phys. Rev. E **101**, 023304 (2020).
8. D. Wu, R. Rossi & G. Carleo, *Unbiased Monte Carlo cluster updates with autoregressive neural networks*, Phys. Rev. Research **3**, L042024 (2021).
9. W. E & E. Vanden-Eijnden, *Towards a theory of transition paths / transition-path theory (committor formalism)*, J. Stat. Phys. **123**, 503 (2006).
10. J. Spanier & E. M. Gelbard, *Monte Carlo Principles and Neutron Transport Problems* (1969) — classical zero-variance / importance theory.